DERIVED FUNCTORS ON BOUNDED AND UNBOUNDED HOMOTOPY CATEGORIES AGREE ON THE BOUNDED PART

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5452-5472, following the statement at 5445-5482.

Editorial note. The passage from cofinality to (1) is the standard colimit-over-cofinal-subcategory identification, flagged in the notes; parts (3) and (4) are the dual mirror.

Allowed standard prerequisites (editorial).

- Quasi-isomorphisms into bounded-below complexes. lemma-complex-cohomology-bounded (same chapter, cited at ~5456). Provides X' -> X" bounded below after any quasi-isomorphism.

Source reference. Original lemma 05RV: sources/stacks-derived/pinned-original-source.tex, source lines 3977–3977. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05T4: sources/stacks-derived/pinned-original-source.tex, source lines 5410–5410. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 05T4.*

(1) *Let X be an object of $K^+(\mathcal{A})$. The right derived functor of $K(\mathcal{A}) \to D(\mathcal{B})$ is defined at X if and only if the right derived functor of $K^+(\mathcal{A}) \to D^+(\mathcal{B})$ is defined at X. Moreover, the values are canonically isomorphic.*
(2) *Let X be an object of $K^+(\mathcal{A})$. Then X computes the right derived functor of $K(\mathcal{A}) \to D(\mathcal{B})$ if and only if X computes the right derived functor of $K^+(\mathcal{A}) \to D^+(\mathcal{B})$.*
(3) *Let X be an object of $K^-(\mathcal{A})$. The left derived functor of $K(\mathcal{A}) \to D(\mathcal{B})$ is defined at X if and only if the left derived functor of $K^-(\mathcal{A}) \to D^-(\mathcal{B})$ is defined at X. Moreover, the values are canonically isomorphic.*
(4) *Let X be an object of $K^-(\mathcal{A})$. Then X computes the left derived functor of $K(\mathcal{A}) \to D(\mathcal{B})$ if and only if X computes the left derived functor of $K^-(\mathcal{A}) \to D^-(\mathcal{B})$.*

Proof. Let X be an object of $K^+(\mathcal{A})$. Consider a quasi-isomorphism $s : X \to X'$ in $K(\mathcal{A})$. By Lemma 05RV there exists quasi-isomorphism $X' \to X''$ with X''

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bounded below. Hence we see that $X/\mathrm{Qis}^+(\mathcal{A})$ is cofinal in $X/\mathrm{Qis}(\mathcal{A})$. Thus it is clear that (1) holds. Part (2) follows directly from part (1). Parts (3) and (4) are dual to parts (1) and (2). $\square$